

PROPOSAL

for

SEMESTER System Curriculum for B.Sc. (Ag.)

As Per:

***Vth DEANS' Committee
ICAR, New Delhi***

Semester wise Course Distribution

Semester-I

Sl. No.	Subject Name	Course Code	Course Title	Credit Hours
1	Agronomy	AGR-101	Fundamentals of Agronomy	4(3+1)
2	Agronomy	AGR-102	Agriculture Heritage*	1(1+0)
3	English	ENG-101	Comprehension and Communication Skills in English	2(1+1)
4	Agricultural Extension	EXT-101	Rural Sociology and Educational Psychology	2(2+0)
5	Agricultural Extension	EXT-102	Human Values and Ethics**	1(1+0)
6	Genetics & Plant Breeding	GPB-101	Fundamentals of Genetics	3(2+1)
7	Horticulture	HOR-101	Fundamentals of Horticulture	2(1+1)
8	Soil Conservation	SCW-101	Introductory Agrometeorology and Climate Change	2(1+1)
9	Soil Science & Agricultural Chemistry	SSC-101	Fundamentals of Soil Science	3(2+1)
10	Biology/Math	BIO-101/ MAT-101	Introductory Biology*/ Elementary Mathematics*	2(1+1)
11	NSS/NCC/ Physical Education	NSS/NCC/ PHYEDU-101	NSS/NCC/Physical Education**	2(0+2)
			Total	24

* - Remedial

** - Non Gradual

Semester- Ist

Course Title: Fundamentals of Agronomy

Course Code: AGR-101

Credit Hours:: 4(3+1)

Theory

Agronomy and its scope, seeds and sowing, tillage and tilth, crop density and geometry, Crop nutrition, manures and fertilizers, nutrient use efficiency, water resources, soil-plant-water relationship, crop water requirement, water use efficiency, irrigation- scheduling criteria and methods, quality of irrigation water, logging.

Weeds- importance, classification, crop weed competition, concepts of weed management principles and methods, herbicides- classification, selectivity and resistance, allelopathy. Growth and development of crops, factors affecting growth and development, plant ideotypes, crop rotation and its principles, adaptation and distribution of crops, crop management technologies in problematic areas, harvesting and threshing of crops.

Practical

Identification of crops, seeds, fertilizers, pesticides and tillage implements, study of agroclimatic zones of India, Identification of weeds in crops, Methods of herbicide and fertilizer application, Study of yield contributing characters and yield estimation, Seed germination and viability test, Numerical exercises on fertilizer requirement, plant population, herbicides and water requirement, Use of tillage implements-reversible plough, one way plough, harrow, leveler, seed drill, Study of soil moisture measuring devices, Measurement of field capacity, bulk density and infiltration rate, Measurement of irrigation water.

Course Title: Agricultural Heritage

Course code: AGR-102

Credit Hours:: 1 (1+0)

Theory

Introduction of Indian agricultural heritage; Ancient agricultural practices, Relevance of heritage to present day agriculture; Past and present status of agriculture and farmers in society; Journey of Indian agriculture and its development from past to modern era; Plant production and protection through indigenous traditional knowledge; Crop voyage in India and world; Agriculture scope; Importance of agriculture and agricultural resources available in India; Crop significance and classifications; National agriculture setup in India; Current scenario of Indian agriculture; Indian agricultural concerns and future prospects.

Practical

Morphology of flowering plants – root, stem and leaf and their modifications. Inflorescence, flower and fruits. Cell, tissues & cell division. Internal structure of root, stem and leaf. Study of specimens and slides. Description of plants - Brassicaceae, Fabaceae and Poaceae.

Course Title: Comprehension and Communication Skills in English

Course Code: ENG-101

Credit Hours: 2(1+1)

Theory

Reading Comprehension, Vocabulary- Antonym, Synonym, Homophones, Homonyms, often confused words. Exercises to Help the students in the enrichment of vocabulary based on competitive examinations. Functional grammar: Articles, Prepositions, Verb, Subject verb Agreement, Transformation, Synthesis, Direct and Indirect Narration. Written Skills: Paragraph writing, Precise writing, Report writing and Proposal writing. The Style: Importance of professional writing. Preparation of Curriculum Vitae and Job applications. Synopsis Writing. Interviews: kinds, Importance and process.

Practical

Listening Comprehension: Listening to short talks lectures, speeches (scientific, commercial and general in nature). Oral Communication: Phonetics, stress and intonation, Conversation practice. Conversation: rate of speech, clarity of voice, speaking and Listening, politeness & Reading skills: reading dialogues, rapid reading, intensive reading, improving reading skills. Mock Interviews: testing initiative, team spirit, leadership, intellectual ability. Group Discussions.

Course Title: Rural Sociology & Educational Psychology

Course Code: EXT-101

Credit Hours: 2(2+0)

Theory

Sociology and Rural sociology: Definition and scope, its significance in agriculture extension, Social Ecology, Rural society, Social Groups, Social Stratification, Culture concept, Social Institution, Social Change & Development. Educational psychology: Meaning & its importance in agriculture extension. Behavior: Cognitive, affective, psychomotor domain, Personality, Learning, Motivation, Theories of Motivation, Intelligence.

Course Title: Human Value and Ethics

Course Code: EXT-102
CreditHours:: 1(1+0)

Theory

Values and Ethics-An Introduction. Goal and Mission of Life. Vision of Life. Principles and Philosophy. Self Exploration. Self Awareness. Self Satisfaction. Decision Making. Motivation. Sensitivity. Success. Selfless Service. Case Study of Ethical Lives. Positive Spirit. Body, Mind and Soul. Attachment and Detachment. Spirituality Quotient. Examination.

Course Title: Fundamentals of Genetics
Course Code: GPB-101
CreditHours:: 3(2+1)

Theory

Pre and Post Mendelian concepts of heredity, Mendelian principles of heredity. Architecture of chromosome; chromonemata, chromosome matrix, chromomeres, centromere, secondary constriction and telomere; special types of chromosomes. Chromosomal theory of inheritance-cell cycle and cell division- mitosis and meiosis. Probability and Chi-square.

Dominance relationships, Epistatic interactions with example. Multiple alleles, pleiotropism and pseudoalleles, Sex determination and sex linkage, sex limited and sex influenced traits, Blood group genetics, Linkage and its estimation, crossing over mechanisms, chromosome mapping. Structural and numerical variations in chromosome and their implications, Use of haploids, dihaploids and doubled haploids in Genetics. Mutation, classification, Methods of inducing mutations & CIB technique, mutagenic agents and induction of mutation. Qualitative & Quantitative traits, Polygenes and continuous variations, multiple factor hypothesis, Cytoplasmic inheritance. Genetic disorders. Nature, structure & replication of genetic material. Protein synthesis.

Practical

Study of microscope.

Study of cell structure.

Mitosis and Meiosis cell division.

Experiments on monohybrid, dihybrid, trihybrid, test cross and back cross,

Experiments on epistatic interactions including test cross and back cross, Practice on mitotic and meiotic cell division,

Experiments on probability and Chi-square test.

Determination of linkage and cross-over analysis (through two point test cross and three point test cross data).

Study of models on DNA and RNA structures.

Course Title: Fundamentals of Horticulture

Course Code: HOR-101

CreditHours: 2 (1+1)

Theory

Horticulture - Its definition and branches, importance and scope; horticultural and botanical classification; climate and soil for horticultural crops; Plant propagation-methods and propagating structures; Seed dormancy, Seed germination, principles of orchard establishment; Principles and methods of training and pruning, juvenility and flower bud differentiation; unfruitfulness; pollination, pollinizers and pollinators; fertilization and parthenocarpy; medicinal and aromatic plants; importance of plant bio-regulators in horticulture. Irrigation – methods, Fertilizer application in horticultural crops.

Practical

Identification of garden tools. Identification of horticultural crops. Preparation of seed bed/nursery bed. Practice of sexual and asexual methods of propagation including micro-propagation. Layout and planting of orchard. Training and pruning of fruit trees. Preparation of potting mixture. Fertilizer application in different crops. Visits to commercial nurseries/orchard.

Course Title: Introductory Agro-Meteorology & Climate Change

Course Code: SCW-101

CreditHours:: 2 (1+1)

Theory

1. **Meteorology:** Meaning and scope of agricultural meteorology; climate and weather.
2. **Earth atmosphere:** Its composition, extent and structure; atmospheric weather variables; atmospheric pressure, its variation with height.
3. **Wind:** Types of wind, daily and seasonal variation of wind speed, cyclone, anticyclone, land breeze and sea breeze.
4. **Solar radiation:** Nature and properties of solar radiation, solar constant, depletion of solar radiation, short wave, longwave and thermal radiation, net radiation, albedo; Atmospheric temperature, temperature inversion,
5. **Atmospheric humidity:** Definition, concept of saturation, vapor pressure, process of condensation.
6. **Precipitation:** Process of precipitation, types and forms of precipitation, cloud formation and classification and artificial rainmaking, rainfall, rainfall intensity and hydrological cycle and its components.
7. **Monsoon:** Mechanism and importance in Indian agriculture,

8. **Weather hazards:** Drought, floods, frost, tropical cyclones and extreme weather conditions such as heat-wave and cold-wave, agriculture and weather relations; modifications of crop microclimate, climatic normals for crop and livestock production.
9. **Weather forecasting:** Types of weather forecast and their uses, climate change, climatic variability, global warming, causes of climate change and its impact on regional and national agriculture.
10. **Climatic Zone:** Indian Agro-Climatic Zones.

Practical

1. Visit of Agro-meteorological observatory, site selection of observatory, exposure of instruments and weather data recording.
2. Measurement of total, shortwave and longwave radiation, and its estimation using Planck's intensity law.
3. Measurement of albedo and sunshine duration, computation of Radiation Intensity using BSS.
4. Measurement of maximum and minimum air temperatures, its tabulation, trend and variation analysis.
5. Measurement of soil temperature and computation of soil heat flux.
6. Determination of vapor pressure and relative humidity.
7. Determination of dew point temperature.
8. Measurement of atmospheric pressure and analysis of atmospheric conditions.
9. Measurement of wind speed and wind direction, preparation of wind rose.
10. Measurement, tabulation and analysis of rain.
11. Measurement of open pan evaporation and evapotranspiration.
12. Computation of PET and AET.
13. Study of wind vane and anemometer.
14. Computation of average rainfall.
15. Measurement of rainfall by Rain Gauge

Course Title: Fundamentals of Soil Science

Course Code: SSC-101

CreditHours:: 3(2+1)

Theory

Soil as a natural body, Pedological and edaphological concepts of soil; Soil genesis: soil forming rocks and minerals; weathering, processes and factors of soil formation; Soil Profile, components of soil; Soil physical properties: soil-texture, structure, density and porosity, soil colour, consistence and plasticity; Elementary knowledge of soil taxonomy classification and soils of India; Soil water retention, movement and availability; Soil air, composition, gaseous exchange, problem and plant reaction-pH, soil acidity and alkalinity, buffering, effect of pH on nutrient

availability; soil colloids - inorganic and organic; silicate clays: constitution and properties; sources of charge; ion exchange, cation exchange capacity, base saturation; soil organic matter: composition, properties and its influence on soil properties; humic substances - nature and properties; soil organisms: macro and micro organisms, their beneficial and harmful effects; Soil pollution - behaviour of pesticides and inorganic contaminants, prevention and mitigation of soil pollution.

Practical

Study of soil profile in field. Study of soil sampling tools, collection of representative soil sample, its processing and storage. Study of soil forming rocks and minerals. Determination of soil density, moisture content and porosity. Determination of soil texture by feel and Bouyoucos Methods. Studies of capillary rise phenomenon of water in soil column and water movement in soil. Determination of soil pH and electrical conductivity. Determination of cation exchange capacity of soil. Study of soil map. Determination of soil colour. Demonstration of heat transfer in soil. Estimation of organic matter content of soil. Estimation of CO_3 and HCO_3 in soil water.

Course Title: Introductory Biology

Course Code: BIO-101

Credit Hours: 2(1+1)

Theory

Introduction to the living world, diversity and characteristics of life, origin of life, Evolution and Eugenics. Binomial nomenclature and classification Cell and cell division. Morphology of flowering plants. Seed and seed germination. Plant systematic- viz; Brassicaceae, Fabaceae and Poaceae. Role of animals in agriculture.

Practical

Morphology of flowering plants – root, stem and leaf and their modifications. Inflorescence, flower and fruits. Cell, tissues & cell division. Internal structure of root, stem and leaf. Study of specimens and slides. Description of plants - Brassicaceae, Fabaceae and Poaceae.

Course Title: Elementary Mathematics

Course Code: MAT-101

Credit Hours: 2(1+1)

Theory

Straight lines : Distance formula, section formula (internal and external division), Change of axes (only origin changed), Equation of co-ordinate axes, Equation of lines parallel to axes, Slope-intercept form of equation of line, Slope-point form of equation of line, Two point form of equation of line, Intercept form of equation of line, Normal form of equation of line,

General form of equation of line, Point of intersection of two st. lines, Angles between two st. lines, Parallellines, Perpendicular lines, Angle of bisectors between two lines, Area of triangle and quadrilateral. Circle: Equation of circle whose centre and radius is known, General equation of a circle, Equation of circle passing through three given points, Equation of circle whose diameters is line joining two points (x_1, y_1) & (x_2, y_2) , Tangent and Normal to a given circle at given point (Simple problems), Condition of tangency of a line $y = mx + c$ to the given circle $x^2 + y^2 = a^2$. Differential Calculus: Definition of function, limit and continuity, Simple problems on limit, Simple problems on continuity, Differentiation of x^n , e^x , $\sin x$ & $\cos x$ from first principle, Derivatives of sum, difference, product and quotient of two functions, Differentiation of functions of functions (Simple problem based on it), Logarithmic differentiation (Simple problem based on it), Differentiation by substitution method and simple problems based on it, Differentiation of Inverse Trigonometric functions. Maxima and Minima of the functions of the form $y = f(x)$ (Simple problems based on it).

Integral Calculus : Integration of simple functions, Integration of Product of two functions, Integration by substitution method, Definite Integral (simple problems based on it), Area under simple well-known curves (simple problems based on it).

Matrices and Determinants: Definition of Matrices, Addition, Subtraction, Multiplication, Transpose and Inverse up to 3rd order, Properties of determinants up to 3rd order and their evaluation.

Practical

- Problems on Addition-Subtraction-Multiplication-Transpose of matrix 3rd order.
- Problems on minor-Cofactor of matrix-Inverse of matrix up to 3rd order.
- Cramer's rule and simple problems based on it and problems on determinants.
- Function limit and continuity with simple problems.
- Problems on differentiation of x^n , e^x , $\sin x$ and $\cos x$ by first principle.
- Derivatives of sum – difference product and quotient of two functions –
- Differentiation of functions of functions-Simple problems based on it.
- Logarithmic-Inverse-Trigonometric functions-Functions of functions
- Equations of tangent-Normal of given curve at given point-Simple problems.
- Problems on integration of functions-Integration of product of two functions-
- Integration by substitution method.
- Integral (simple problems based on it)-Area under simple well-known curves.
- Problems on different types of straight line forms.